

Nishal Thomas

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Education

St. Mary's Sr. Sec. School

Class 12

New Delhi, India

Ongoing

Class 11 — Subject Topper: Computer Science, Chemistry, and English;

2026

2nd in Mathematics (Science stream, cohort of approximately 100)

Class 10 [CBSE] — 91%

2025

Work Experience

Deskflow

[\[deskflow.odddworks.org\]](https://deskflow.odddworks.org)

Founder

May 2026 – Present

- Founded Deskflow, an AI voice-agent platform that automatically answers and manages incoming clinic phone calls when front-desk staff are unavailable, addressing the missed-call problem for small clinics.
- Built and piloted the product across dental and ENT clinics in Jaipur and Delhi, enabling live, calendar-verified appointment booking, rescheduling, and cancellation in Hindi, Hinglish, and English.
- Designed the assistant's real-time, calendar-verified booking logic and call-escalation rules, ensuring it only confirms appointments that exist in the clinic's diary and defers medical questions directly to staff.

Research

FormInv: A Measurement Protocol for Semantic Invariance in Mathematical Reasoning Benchmarks [\[arXiv\]](#)

[First Author • ICML 2026 AI4Math Workshop (Poster)]

- First author of an accepted ICML 2026 workshop paper introducing FormInv, a benchmark and evaluation framework for measuring semantic invariance in mathematical reasoning systems
- Developed a dataset of Lean4-grounded theorem paraphrases spanning 8 transformation families and evaluated robustness across 9 frontier language models
- Proposed Semantic Consistency Rate (SCR) and cross-model unanimity auditing, revealing formulation-sensitive failure modes and benchmark construction errors hidden by aggregate accuracy metrics

When Models Know Without Representing: Behavioural–Representational Dissociation in LLM Abstraction Hierarchies [\[Under Review\]](#) [\[Code\]](#)

- Constructed 50 abstraction ladders spanning nine levels of conceptual abstraction across mathematics and scientific domains
- Investigated behavioural–representational dissociation in language models using representation geometry, abstraction coherence metrics, and activation patching
- Conducted experiments across model scales from 70M to 70B parameters to study abstraction formation and scaling behaviour

Mapping Hallucinations in Tiny Aya

Multilingual Hallucination Detection Study

- Participated in Expedition Tiny Aya, a global research initiative by Cohere Labs, collaborating within a five-member international team to investigate multilingual hallucinations in Tiny Aya models
- Applied black-box metrics (Hallucination Rate, Self-Consistency, Variance, Cross-Model Disagreement, Prompt Sensitivity Score, etc.) alongside white-box methods to detect hallucination signals using Python-based pipelines

Technical Projects

Step-Free Arithmetic Transformers

Mechanistic Interpretability Study

- Built a reproducible research framework for training compact arithmetic transformers (0.66M–10M parameters) on hierarchical mathematical reasoning tasks using PyTorch and mixed-precision training.
- Applied mechanistic interpretability techniques including attention-head ablation, linear probing, and out-of-distribution evaluation to identify parenthesis-sensitive computation circuits underlying arithmetic reasoning.

Aletheia

Evidence-Grounded LLM Assistant

- Engineered a transparent retrieval-augmented reasoning pipeline using techniques from **Loki** (claim decomposition),

LM-Polygraph (verbalized confidence estimation), **FIRE** (adaptive retrieval), and **OpenFactCheck** (evidence-grounded verification), integrating them into a unified FastAPI, Qdrant, Ollama, and Next.js system.

- Developed a self-enriching RAG architecture that automatically retrieves and indexes arXiv and Wikipedia content when retrieval coverage is insufficient, enabling continual expansion of the local knowledge base.

Grokking Feasibility Boundaries in Transformers

Generalization Dynamics Study

- Constructed controlled interpolation–extrapolation benchmarks for modular arithmetic to isolate the effect of train–test overlap on grokking behaviour in compact transformer models.
- Investigated delayed generalization through representation geometry, representation rank collapse, and attention entropy, identifying empirical boundaries for grokking under varying supervision regimes.

Continuum [K2-Think Hackathon]

AI Safety Infrastructure

- Designed an AI safety system for detecting epistemic drift and contradiction accumulation across long-running LLM conversations through cumulative stance tracking and dependency analysis.
- Implemented a hybrid verification pipeline combining symbolic heuristics with external reasoning models to identify, validate, and visualize conversational inconsistencies.

Extracurriculars

- Participated in multiple speaking competitions including speech and debate, securing awards and recognitions at inter-school level.
- Served as a member of the School Student Council, contributing to student representation and initiatives
- Qualified Stage 1 of the Indian National Olympiad of Artificial Intelligence (INAIO)
- Completed the CBSE–Expressions India Adolescent Peer Educator’s Leadership Program in holistic development and well-being

Awards

2025: YMCA Academic Excellence Award — Class 10 Board performance

2025: Science Topper — Class 10, St. Mary’s Sr. Sec. School

2025: 100/100 in Artificial Intelligence — CBSE Board Examination